A PRESENTATION OF THE BLOWUP ALGEBRA OF AN H1-REGULAR SEQUENCE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A regular sequence is Koszul regular and a Koszul regular sequence is H1-regular. Lemmas \ref{lemma-regular-koszul-regular} and \ref{lemma-koszul-regular-H1-regular}, cited at proof lines 7885-7887. Explain why the hypothesis of the lemma covers the classical cases of regular and Koszul regular sequences.
- The blowup algebra is the quotient of the polynomial ring over R by the ideal of torsion elements. Algebra, Lemma \ref{algebra-lemma-affine-blowup-quotient-description}, cited at proof line 7894. Reduces the identification of the blowup algebra to the absence of torsion by the first element.
- Koszul complexes are functorial under an isomorphism of the underlying modules, and Koszul regularity is preserved by flat base change. Lemma \ref{lemma-functorial} and Lemma \ref{lemma-koszul-regular-flat-base-change}, cited at proof lines 7915-7917. Transfer H1-regularity from the given sequence to the transformed sequence in the polynomial ring.
- The Koszul complex of a, ay_2 - a_2, ..., ay_r - a_r is the cone on multiplication by a on the Koszul complex of the remaining elements. Lemma \ref{lemma-cone-koszul}, cited at proof lines 7918-7919. Turns vanishing of the first homology of the Koszul complex into injectivity of multiplication by a on the zeroth homology.

Source reference. Original lemma 0BIQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14498–14498. The lemma's own canonical tag,

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quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 062F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7794–7794. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0CEM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15582–15582. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0G8S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20277–20277. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0624: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7783–7783. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 062H: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7796–7796. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0629: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7788–7788. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $a, a_2, \ldots, a_r$ be an H_1-regular sequence in a ring R (for example a Koszul regular sequence or a regular sequence, see Lemmas 062F and 0CEM). With $I = (a, a_2, \ldots, a_r)$ the blowup algebra $R' = R[\frac{I}{a}]$ is isomorphic to $R'' = R[y_2, \ldots, y_r]/(ay_i - a_i)$.*

Proof. By Algebra, Lemma 0G8S it suffices to show that R'' is a-torsion free.

We claim $a, ay_2 - a_2, \ldots, ay_n - a_r$ is a H_1-regular sequence in $R[y_2, \ldots, y_r]$. Namely, the map

$$(a, ay_2 - a_2, \ldots, ay_n - a_r) : R[y_2, \ldots, y_r]^{\oplus r} \longrightarrow R[y_2, \ldots, y_r]$$

used to define the Koszul complex on $a, ay_2 - a_2, \ldots, ay_n - a_r$ is isomorphic to the map

$$(a, a_2, \ldots, a_r) : R[y_2, \ldots, y_r]^{\oplus r} \longrightarrow R[y_2, \ldots, y_r]$$

used to the define the Koszul complex on $a, a_2, \ldots, a_r$ via the isomorphism

$$R[y_2, \ldots, y_r]^{\oplus r} \longrightarrow R[y_2, \ldots, y_r]^{\oplus r}$$

sending $(b_1, \ldots, b_r)$ to $(b_1 - b_2y_2 \ldots - b_ry_r, -b_2, \ldots, -b_r)$. By Lemma 0624 these Koszul complexes are isomorphic. By Lemma 062H applied to the flat ring map $R \to R[y_2, \ldots, y_r]$ we conclude our claim is true. By Lemma 0629 we see that the Koszul complex K on $a, ay_2 - a_2, \ldots, ay_n - a_r$ is the cone on $a : L \to L$ where L is the Koszul complex on $ay_2 - a_2, \ldots, ay_n - a_r$. Since $H_1(K) = 0$ by the claim, we conclude that $a : H_0(L) \to H_0(L)$ is injective, in other words that $R'' = R[y_2, \ldots, y_r]/(ay_i - a_i)$ has no nonzero a-torsion elements as desired. □